

Data Engineering and Decision Systems Engineer

SARRA BEN THAYER

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PROFILE

A Data Engineering graduate with hands-on experience in data pipeline design, database systems, and data modeling, gained through academic projects and internships . Diligent and detail-oriented, with a genuine drive to learn and take on new challenges. Passionate about transforming raw data into clear, actionable insights that drive real-world impact, and seeking an opportunity to further develop technical expertise within a data-driven team

EDUCATION

Data Engineering and Decisional Systems : National School of Electronics and Telecommunications of Sfax 2023 – 2026

Physics-Technology Preparatory Cycle : Preparatory Institute for Engineering Studies of Nabel 2021 – 2023

PROFESSIONAL EXPERIENCE

AI-Powered Data Modeler — End-of-Studies Project —TALAN 2/2026 - 6/2026

- Built a multi-agent AI system automating data modeling from raw files (CSV, Excel, PDF, text) to deployed database schemas
- Parser Agent: Extracts structured schema elements (tables, columns, primary/foreign keys, data types) directly from heterogeneous raw documents using LLM-based parsing, then persists the normalized schema to PostgreSQL for downstream processing.
- Extractor Agent: Transforms the extracted schema into a semantic knowledge graph that captures entity relationships, dependencies, and business semantics beyond raw structure; every graph state is versioned in Neo4j, enabling full traceability and rollback
- Validator Agent: Runs automated structural checks (referential integrity, key consistency) and semantic checks (relationship coherence, naming/type consistency) against the knowledge graph, producing a detailed validation report alongside an auto-corrected graph reducing human review time on data quality issues.
- Modeler Agent:Converts the validated graph into formal data modeling schemas (Star, Snowflake, Galaxy), deploys them automatically, and exposes a natural language query interface so non-technical users can query the resulting model directly.
- Incremental Update Pipeline : Uses content hashing to detect schema drift between successive file versions, then auto-generates the corresponding migration scripts avoiding full re-modeling on every data update

Banking Chatbot — Summer Intern — Société Tunisienne de Banque (STB) 6/2025 - 7/2025

- Built an NLP-powered chatbot for handling client account queries and financial insights by training custom intent recognition models for natural language understanding.
- Developed an interactive React interface enabling real-time conversational interactions

ACADEMIC PROJECTS

Real-Time Social Media Analytics Platform

- Designed a streaming pipeline ingesting social media posts in real time via Kafka
- Processed streams with Spark for sentiment analysis, hashtag tracking, and time-windowed statistics
- Used Redis to cache frequent/real-time stats, reducing direct PostgreSQL load
- Stored analysis results (sentiment scores, hashtag counts, trend stats) in PostgreSQL
- Built backend APIs with Flask, connecting processing logic to an interactive dashboard
- Containerized services with Docker for simplified orchestration and deployment

Real-Time Sales Streaming

- Built an end-to-end streaming architecture with Apache Kafka to ingest simulated real-time sales data, and Spark Structured Streaming for micro-batch processing with watermarking to handle late-arriving events
- Implemented a modern Lakehouse with Delta Lake following the medallion architecture (Bronze → Silver), ensuring ACID transactions, time-travel capabilities, and incremental aggregations for business analytics
- Developed a complete Python/PySpark pipeline: Kafka producer simulating realistic sales transactions, Spark consumers for real-time transformations, and Power BI dashboard connected to Silver tables for KPI visualization

MLOps Pipeline Project

- Built an end-to-end MLOps pipeline for an XGBoost crime-prediction model, with CI/CD in Jenkins triggered on every Git push.
- Versioned data with DVC and DagsHub; tracked experiments and registered models in MLflow; validated models with Deepchecks.
- Served the model via FastAPI and Streamlit, containerized with Docker and deployed on Kubernetes (load balancing, rolling updates).
- Monitored data drift and model performance with Evidently AI, automatically triggering retraining when drift is detected.

Technical Skills

- **Artificial Intelligence**: ML, DL, LLMs, NLP, MLOps, Generative AI, LSTM, Knowledge Graphs
- **Programming Languages**: Python, C, Java, Scala, R
- **Big Data Technologies:** Apache Spark, PySpark, Spark Streaming, Kafka, DataBricks, Hive, HDFS, Hadoop
- **Databases**: SQL, NoSQL, MySQL, MongoDB, Neo4j, PostgreSQL, DBeaver , Vector Data bases
- **Frameworks and Tools**: REST API, PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, LangGraph, LangChain, Power BI
- **Containerization**: Docker, Kubernetes

Langues

Arabic : Native **English :** Fluent **French :** Fluent

Extracurricular Activities

- IEEE Computer Society Vice Chair 2023 – 2024
- AIESEC member - Outgoing Global Volunteer 2021 – 2023